

```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('default')
```

```
df =pd.read_csv("data.csv")
```

```
df.head()
```

	id	gender	age	hypertension	heart_disease	ever_married	\
0	9046	Male	67.0	0	1	Yes	
1	51676	Female	61.0	0	0	Yes	
2	31112	Male	80.0	0	1	Yes	
3	60182	Female	49.0	0	0	Yes	
4	1665	Female	79.0	1	0	Yes	

	work_type	Residence_type	avg_glucose_level	bmi	
0	Private	Urban	228.69	36.6	formerly smoked
1	Self-employed	Rural	202.21	NaN	never smoked
2	Private	Rural	105.92	32.5	never smoked
3	Private	Urban	171.23	34.4	smokes
4	Self-employed	Rural	174.12	24.0	never smoked

	stroke
0	1
1	1
2	1
3	1
4	1

```
print("THE ROWS AND COLUMNS OF DATA FRAME")
df.shape
```

```
THE ROWS AND COLUMNS OF DATA FRAME
```

```
(5110, 12)
```

```
print("THE BASIC INFORMATION ABOUT DATA FRAME")
print(df.info())
```

```
THE BASIC INFORMATION ABOUT DATA FRAME
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 5110 entries, 0 to 5109

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	id	5110 non-null	int64
1	gender	5110 non-null	object
2	age	5110 non-null	float64
3	hypertension	5110 non-null	int64
4	heart_disease	5110 non-null	int64
5	ever_married	5110 non-null	object
6	work_type	5110 non-null	object
7	Residence_type	5110 non-null	object
8	avg_glucose_level	5110 non-null	float64
9	bmi	4909 non-null	float64
10	smoking_status	5110 non-null	object
11	stroke	5110 non-null	int64

dtypes: float64(3), int64(4), object(5)

memory usage: 479.2+ KB

None

```
print("TOTAL NO.OF UNIQUE VAULES")
```

```
print(df.id.unique())
```

TOTAL NO.OF UNIQUE VAULES

[9046 51676 31112 ... 19723 37544 44679]

```
print("SEARCHING FOR DUPLICATE VALUES")
```

```
df.duplicated().sum()
```

SEARCHING FOR DUPLICATE VALUES

```
np.int64(0)
```

```
df.describe()
```

	id	age	hypertension	heart_disease	\
count	5110.000000	5110.000000	5110.000000	5110.000000	
mean	36517.829354	43.226614	0.097456	0.054012	
std	21161.721625	22.612647	0.296607	0.226063	
min	67.000000	0.080000	0.000000	0.000000	
25%	17741.250000	25.000000	0.000000	0.000000	
50%	36932.000000	45.000000	0.000000	0.000000	
75%	54682.000000	61.000000	0.000000	0.000000	
max	72940.000000	82.000000	1.000000	1.000000	

	avg_glucose_level	bmi	stroke
count	5110.000000	4909.000000	5110.000000
mean	106.147677	28.893237	0.048728
std	45.283560	7.854067	0.215320
min	55.120000	10.300000	0.000000
25%	77.245000	23.500000	0.000000

50%	91.885000	28.100000	0.000000
75%	114.090000	33.100000	0.000000
max	271.740000	97.600000	1.000000

```
print("TOTAL NO.OF NULL VALUES")
```

```
print(df.isnull().sum())
```

```
TOTAL NO.OF NULL VALUES
```

```
id          0
gender      0
age         0
hypertension 0
heart_disease 0
ever_married 0
work_type   0
Residence_type 0
avg_glucose_level 0
bmi         201
smoking_status 0
stroke      0
```

```
dtype: int64
```

```
print(df.groupby('gender')['stroke'].mean())
```

```
gender
```

```
Female    0.047094
```

```
Male      0.051064
```

```
Other     0.000000
```

```
Name: stroke, dtype: float64
```

```
print(df.groupby('work_type')['stroke'].mean())
```

```
work_type
```

```
Govt_job    0.050228
```

```
Never_worked 0.000000
```

```
Private     0.050940
```

```
Self-employed 0.079365
```

```
children    0.002911
```

```
Name: stroke, dtype: float64
```

```
print(df.groupby('Residence_type')['stroke'].mean())
```

```
Residence_type
```

```
Rural      0.045346
```

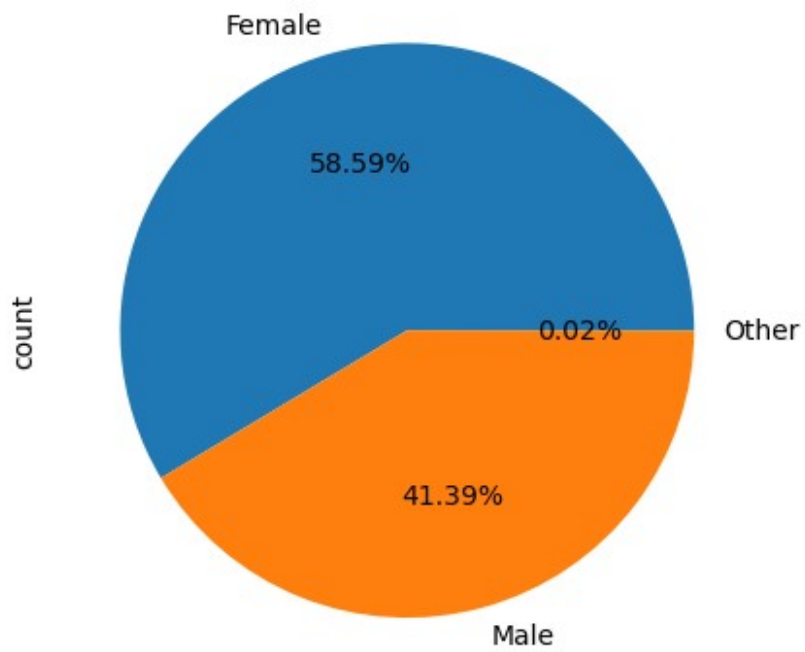
```
Urban      0.052003
```

```
Name: stroke, dtype: float64
```

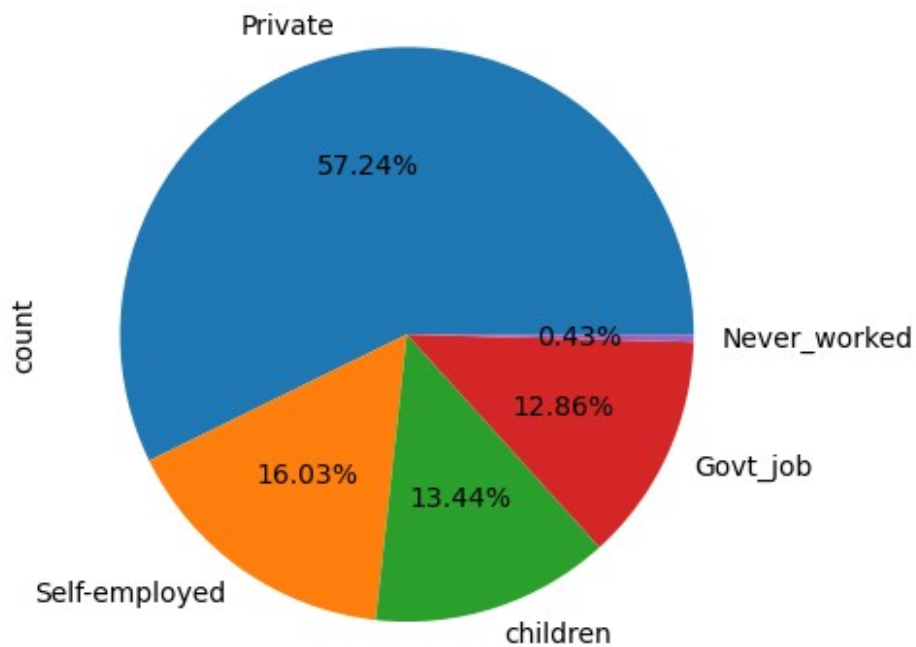
```
(5110, 12)
```

```
df["gender"].value_counts().plot(kind="pie", autopct='%1.2f%%')
```

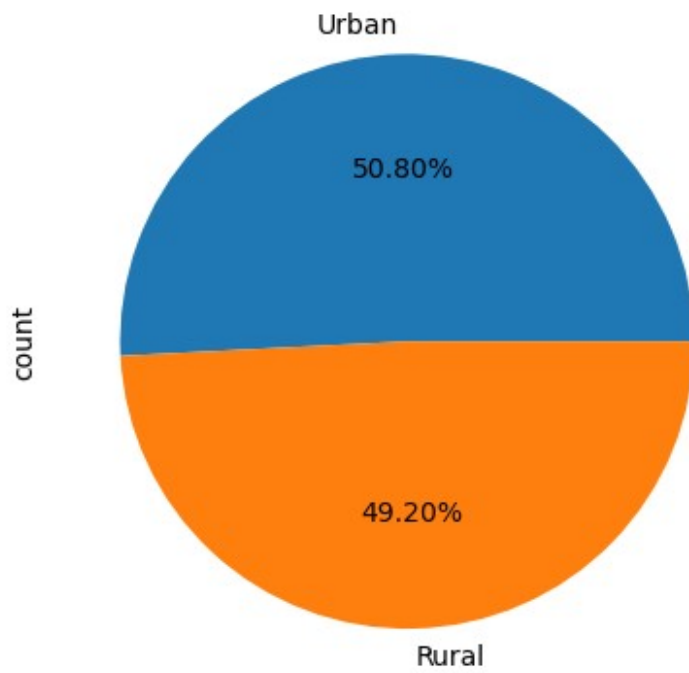
```
<Axes: ylabel='count'>
```



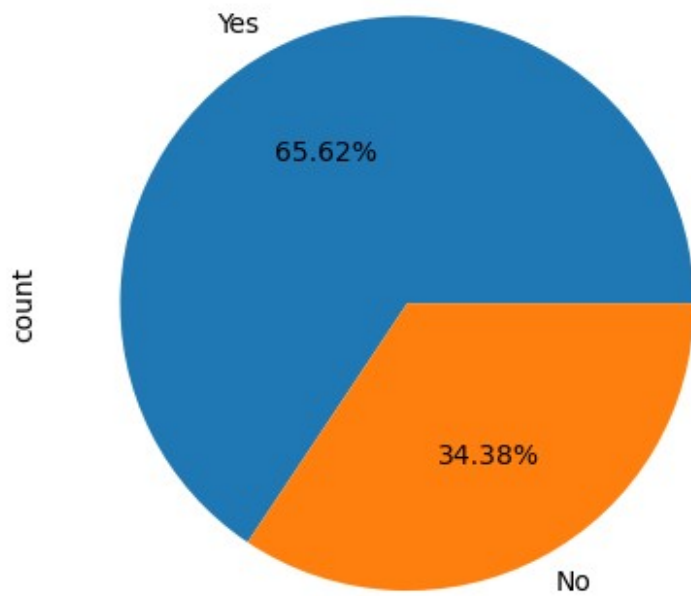
```
print("WORK TYPE OF PATIENTS")
df["work_type"].value_counts().plot(kind="pie", autopct='%1.2f%%')
WORK TYPE OF PATIENTS
<Axes: ylabel='count'>
```



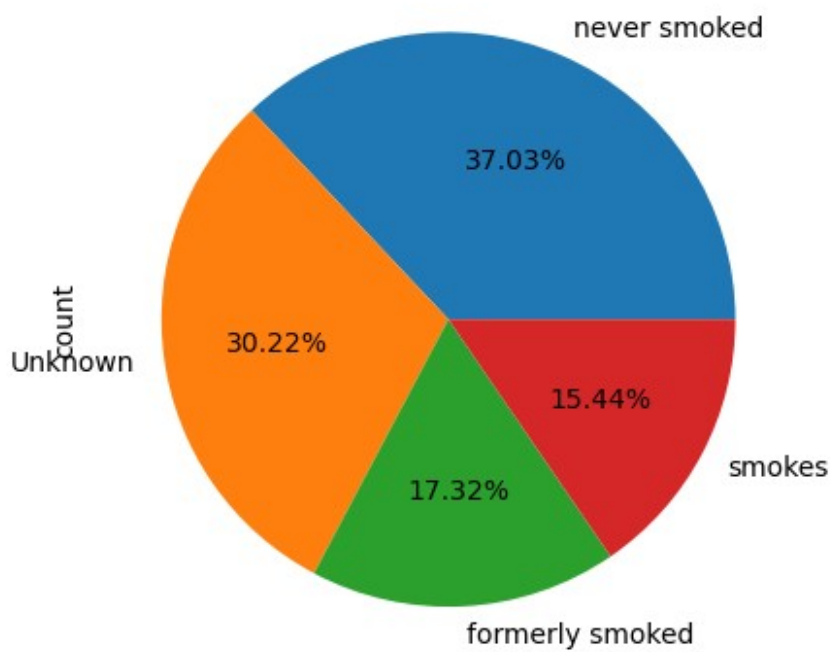
```
print("PATIENT'S RESIDENTIAL AREA ")
df["Residence_type"].value_counts().plot(kind="pie", autopct='%1.2f%%')
PATIENT'S RESIDENTIAL AREA
<Axes: ylabel='count'>
```



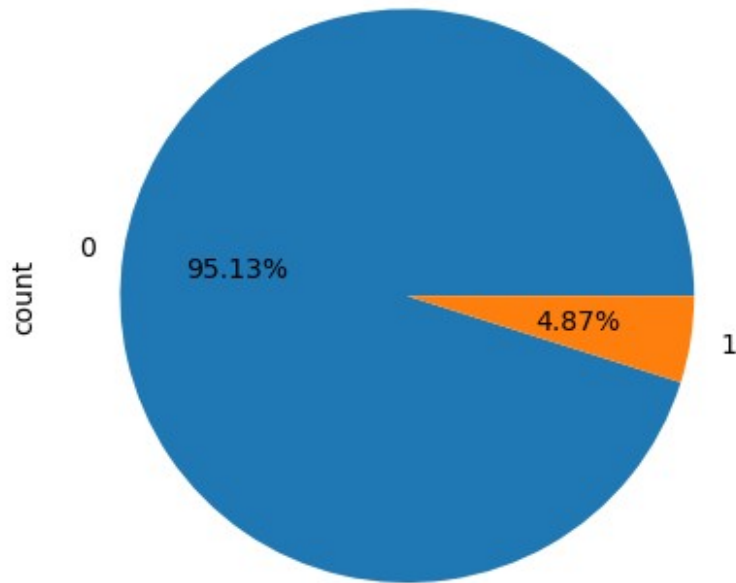
```
df.heart_disease.sum()
276
df.hypertension.sum()
498
df.heart_disease.sum()
276
df["ever_married"].value_counts().plot(kind="pie", autopct='%1.2f%%')
<Axes: ylabel='count'>
```



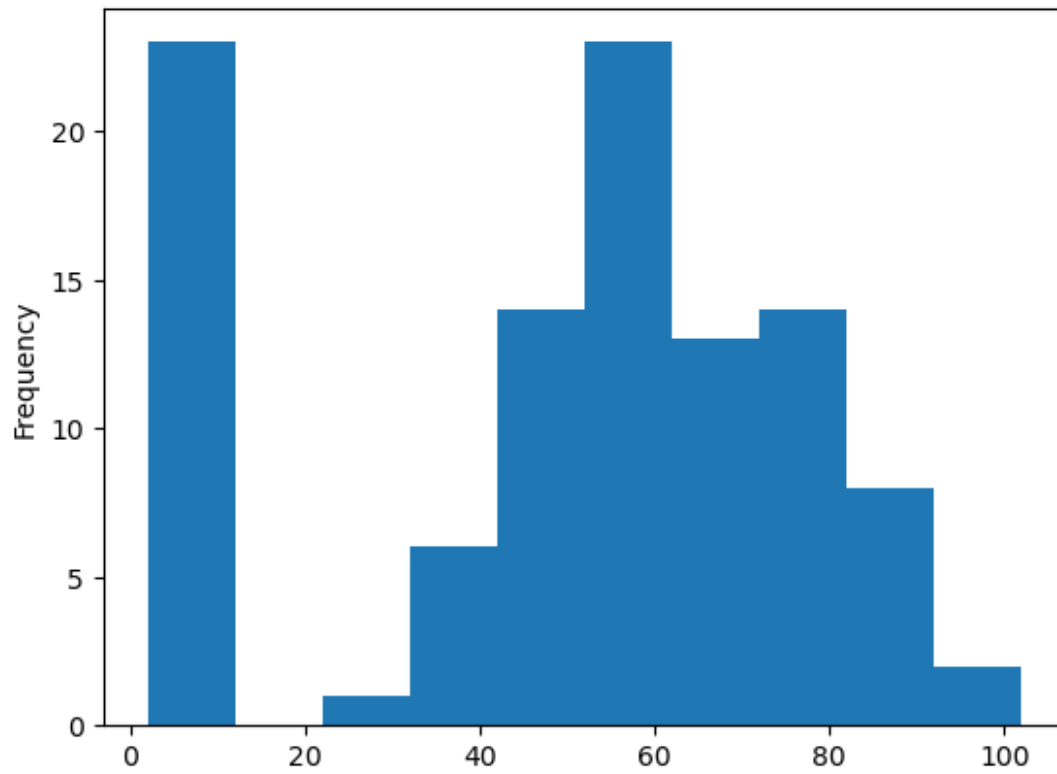
```
df["smoking_status"].value_counts().plot(kind="pie", autopct='%1.2f%%')  
<Axes: ylabel='count'>
```



```
df["stroke"].value_counts().plot(kind="pie", autopct='%1.2f%%')  
<Axes: ylabel='count'>
```



```
df["age"].value_counts().plot(kind="hist")  
<Axes: ylabel='Frequency'>
```

```
df["bmi"].value_counts().plot(kind="hist")  
<Axes: ylabel='Frequency'>
```

